Air Quality Planning for Outdoor Events

Final project proposal

We propose a cloud-based analytics application for an outdoor activity operator managing fitness classes, sports camps, or walking tours. The project will compare historical PM2.5 exposure across locations and session times and display current official air-quality information.

Who is on the team

Team member	Primary responsibility
Yujie Chen	Source access and cloud ingestion of historical and live data
Breanna Young	Normalized operational store and data quality reconciliation
Bravo Nie	Snowflake warehouse and exploratory and comparative analysis
Tongyan Liu	Streamlit application and Claude access through an MCP server

We will work in three two-week sprints: source access and ingestion; warehouse and validated analysis; application and demonstration. Each sprint will have a shared goal, peer review, and an integrated demonstration. Primary owners remain accountable for their stages, and every member will present their contribution. Final integration and submission preparation will follow the third sprint.

What problem are we solving

An outdoor activity operator must choose locations and session times before knowing whether air quality will disrupt programming. A city with a favorable average may still experience prolonged pollution episodes, while morning and afternoon conditions may differ. These patterns matter when choosing regular schedules and planning contingency capacity.

Our central question is: which locations and recurring session times have historically experienced lower PM2.5 concentrations and fewer prolonged pollution episodes? We will initially compare Chicago, New York, Los Angeles, Seattle, and Denver during 2022–2025. A useful answer will help the operator compare scheduling options and recognize conditions requiring review, without assuming that historical patterns guarantee future conditions.

How do we plan to solve it

We will combine EPA historical observations and site metadata with an ongoing AirNow feed. Google Cloud Storage will retain raw files, Cloud SQL will hold normalized operational records, and Snowflake will support a dimensional warehouse and reconciled analytical queries. We will examine missingness, distributions, outliers, and monitor coverage; compare morning and afternoon concentrations within the same monitor and date; and measure seasonal patterns and consecutive elevated-pollution hours. Streamlit will present the findings and current official AQI, while Claude through an MCP server will explore warehouse results. All pipeline stages will run in the cloud using available free tiers or trial credits.

What data are we using

The project uses three complementary inputs. Public file downloads avoid private data agreements and API-key registration. We will attribute the EPA and contributing monitoring agencies, follow the applicable data-use guidance, and label AirNow observations as preliminary. [1–4]

EPA AirData historical hourly PM2.5 observations. We will download the 2022–2025 annual ZIP/CSV files for parameter 88101 from EPA AirData, then filter to the five selected metropolitan areas. EPA lists approximately 35.36 million rows across these four nationwide files; the final study subset will be smaller. The 2025 download contains 8,367,660 listed rows and is approximately 68.6 MB compressed. Its download endpoint was reachable during our access check. These records support historical comparisons; they are not our standalone big-data qualification. [1]

EPA monitoring-site reference data. The public aqs_sites.zip file contains approximately 21,000 site records nationally. It provides geographic and descriptive context for selecting metropolitan areas and joining observations to locations. The endpoint was reachable during our access check. This small reference input enriches the analysis rather than meeting the big-data rule itself. [1]

AirNow ongoing hourly observations. We will collect the public HourlyAQObs_yyyymmddhh.dat files from the AirNow date directories as new observations arrive. The October 27, 2026, 12:00 UTC sample downloaded successfully: 1,009,817 bytes and 4,416 site records, including records without PM2.5 values. This is one observed file, not a guaranteed hourly volume. AirNow documents hourly publication and revisions to the preceding 72 hours. This feed supplies current official AQI for a clearly labeled live display. [2, 3]

Why this meets the big data rule

We will use the assignment's real-time or streaming pathway. The cloud pipeline will continuously ingest newly published hourly records, incrementally process updates, and refresh the live view. It will retain arrival times, handle revised records, and expose data freshness. We will demonstrate this ongoing ingestion during the project; a static historical download or a manually refreshed dashboard alone will not be our claim to meeting the requirement.

Analytical value and practical scope

The core deliverables are paired comparisons of session times, seasonal distributions by city, and the duration of elevated-concentration episodes under a clearly stated analytical threshold. We will check missing-hour gaps and monitor coverage before interpreting differences, preserve measurement-method distinctions, and follow EPA's PM2.5 correction advisory. Historical findings will use AQS/AirData; preliminary AirNow data will support the current-condition display. We will not infer cancellations or financial savings without operator records. Forecasting and weather enrichment are optional extensions. [1, 3–5]

Sources

[1] EPA AirData downloads

[2] AirNow hourly file specifications

[3] AirNow data exchange guidelines

[4] AirNow public file archive

[5] EPA PM2.5 data advisory